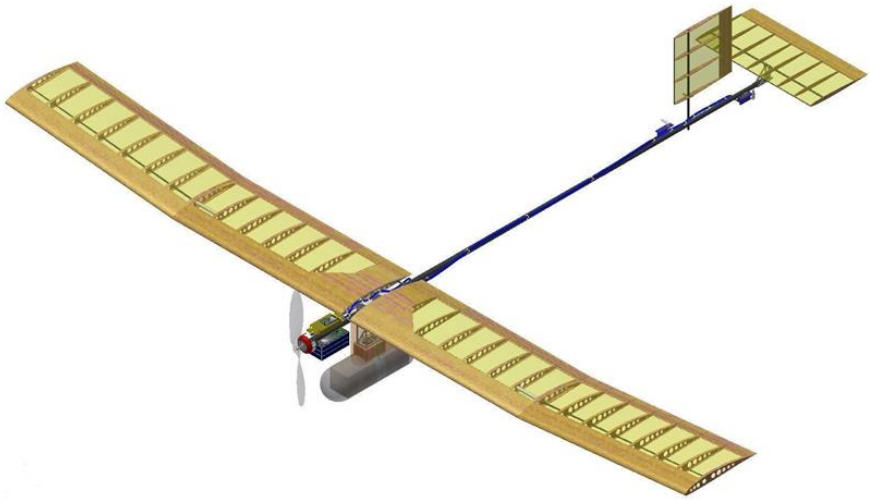
SAE Aero Design East 2015

Academic Flight Club



Pszczola

Team number 328

MICRO CLASS

Poznan University of Technology

2015 SAE AERO DESIGN

STATEMENT OF COMPLIANCE

Certification of Qualification

Team Name	Pszczola	Team Number	328
School	Poznan University of Technology		
Faculty Advisor	Radoslaw Gorzenski		
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Statement of Compliance

As Faculty Advisor, I certify that the registered team members are enrolled in collegiate courses. This team has designed, constructed and/or modified the radio controlled airplane they will use for the SAE Aero Design 2015 competition, without direct assistance from professional engineers, R/C model experts or pilots, or related professionals.

Signature of Faculty Advisor

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List of Symbols and Acronyms

α – angle of attack	Rn – Flight Round Score
AR – Aspect Ratio	S_{ref} – wing area
B_{AD} – Assembly Demonstration Bonus	t – time (assembly time)
B_{OA} – operational availability	v – flow speed
Cd – drag coefficient	V_h, V_v – tail volume coefficients
Cl – lift coefficient	
Cm – pitching moment	
EA – elevator + ailerons	
FFS – Final Flight Score	
L – characteristic length	
L_{AC} – Aircraft Container Length	
L_h, L_v – control surface torque arm lenght	
m – total number of rounds	
MAC – Mean Aerodynamic Chord	
n – number of successful flights	
ν – kinematic viscosity	
PET - Polyethylene terephthalate	
PF – Payload Fraction	
RE – rudder + elevator	
Re – Reynolds number	
REA – rudder + elevator + ailerons	

1. Introduction

1.1 Abstract

The Academic Flight Club, which functions within the Poznan University of Technology in Poland, presents the technical report concerning our submission for the Aero Design East 2015 Micro Class competition: the “Pszczola” aircraft. The report follows the traditional outline of designing: objective and constraint analysis, conceptual design, preliminary design, detail design and design analysis, including testing, aerodynamic calculations, finite elements method of structural analysis and CFD analysis.

The solution we eventually came up with is a high wing monoplane, with a single cargo pod below the fuselage. It is equipped with one brushless electric motor EMAX GT2205/22 with a GWS 8x4 propeller. The airplane has two traditional control surfaces: rudder and elevator, and a wing span of 1415mm with the wind chord of 150 mm. The final design derives from one of our team’s previous model, the “Big Bang” which took part in the SAE Aero Design West competition in 2013.

All 3D and 2D drawings were made using Autodesk Inventor 2014 software.

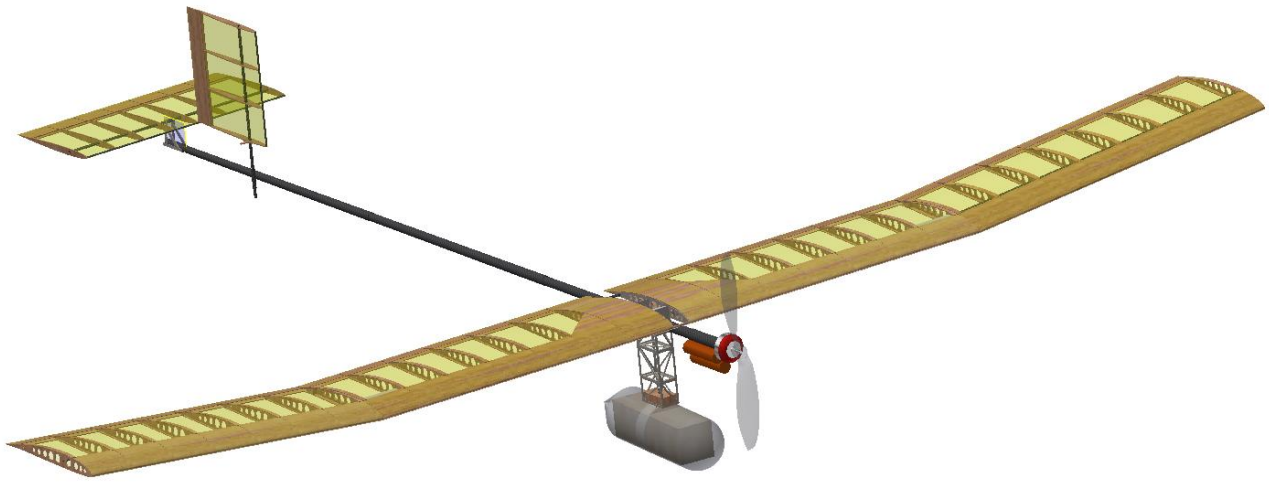


Fig. 1 General outlook on “Pszczola”

1.2 Mission objective

1.2.1 Design objectives and constraints

The idea behind a Micro Class airplane is that it should be a small, light, mobile aircraft, which should be easily carried, and quickly deployed, while maintaining good load capacity and reliability to perform every time it is needed. The disassembled model should fit in a container in a form of a tube, no more than 6” in diameter. A successful flight round is comprised of a hand-launch, approaching a turn point, performing an 180° turn, flying the distance of 300 ft towards the second turn point, performing a second 180° turn and landing within the landing zone.

The Poznan University of Technology Aero Design team took part in earlier editions of the competition, so we compared the previous and current requirements to evaluate in what degree can we rely on preceding solutions. However, the present team is mostly comprised of new members, so to take that into account, we added an additional constraint: limiting the complexity of the vehicle and using construction paradigms similar to the previous ones, in order to make it easier to build for the less experienced, . Cost efficiency was also taken under consideration. Below you will find a table (Fig.2) with a summary of the most important detail requirements, including those imposed by ourselves (marked in blue).

Requirement category	Requirements in 2013	Requirements in 2015	Changes/commentary
Weight and dimensions	•Carrying case measuring 24 x 18 x 8 in. •Payload dimensions: 5x2x2 in. •Lowest empty weight possible – max 200g •Transmitter size: 7.7x7.7x5 in. fits into the box	•Carrying case in shape of a tube, max. ϕ 6”. •Payload dimensions 1.5”x1.5”x5” •The case, containing the aircraft, must not weigh more than 10 lbs. •Lowest possible empty weight for good score	•Payload is smaller, but similar in nature •Certain elements might have to be divided into parts to fit in the case •Minimising the case size will benefit score
Systems	•Electric brushless motor with 100W power	•Electric motor •Max. 3 cell 2200mAh	•Previous model performed well in terms of weight,

	• 400-500g thrust • LI-Pol battery 450mAh • No landing gear	Li-Po battery pack	lift and power – we should adapt a similar configuration • A skid instead of landing gear might benefit the weight
Operation	• Lowest takeoff weight possible • Flying a 360° circuit of the flying field in max 1,5 min • Takeoff by hand or a launching device, landing in 200ft. • Launch by hand produces velocities of 10-12m/s (stall speed) • Reliability factor has important input into final score	• Fast assembly in the 1st round • Hand launch only, landing within a 200 ft zone • 2,4GHz radio control • Launch preparations must not exceed 90 seconds	• Must provide a grip for the hand launch • The construction was farily robust, and scored 3rd by payload fraction • Most requirements are similar
Time/budget/ other	• 3 weeks for manufacturing • $300 budget (for aircraft) • 2000$ budget (for logistics)	• 1 week per airplane for manufacturing (does not concert prototypes) • 500$ budget for aircraft • 2000$ budget for logistics	• The design must be easy to build • Previous budget was appropriate – we have a margin for trial and error • Using old parts as examples of details of construction will be beneficial

Fig. 2 Requirements table

Overall, after initial analysis of the requirements as presented above, we were leaning towards constructing a similar airplane as before. It would be beneficial to be able to use parts the previous construction as a benchmark example to guide new team members in building new parts.

1.3 Work schedule

	Task	Start	End	Nov.	Dec.	Jan.	Feb.	Mar.
I	Design							
1	Brainstorming	1.11.2014	3.11.2014					
2	Preliminary Phase	4.11.2014	6.11.2014					
3	Detailed Phase	7.11.2014	15.11.2014					
4	Final Reviev	16.11.2014	18.11.2014					
5	Report							
6	Draft	30.11.2014	10.12.2014					
7	Final Draft	11.12.2014	16.12.2014					
8	Final Report Due	20.01.2015						
II	Manufacturing							
1	Prototype Aircraft	20.11.2014	10.12.2014					
2	Construction Changes	6.01.2015	12.01.2015					
3	Final Aircraft	15.01.2015	24.01.2015					
III	Testing							
1	Materials	18.11.2015	20.11.2014					
2	Battery	10.12.2014	11.12.2014					
3	Propulsion	15.12.2014	18.12.2015					
4	Prototype Testing	27.12.2014	3.01.2015					
5	Drop Test	4.01.2015	4.01.2015					
6	Flight Test	25.01.2015	26.01.2015					

Fig. 3. Work Schedule

2. Design process

2.1 Research and conceptual design

2.1.1 Score analysis

The scoring for the 2015 Micro Class is derived from the following formulas:

$$FFS = \left(\frac{1}{n}\sum_{1}^{n} Rn\right) + \left(\sum_{1}^{m} B_{OA}\right) + B_{AD}$$

Where the R_n is the score for each flight round

$$R_n = (70 - L_{AC})(2 \cdot PF)$$

Where the L_{AC} is the aircraft container length in inches, PF is the Payload Fraction, defined by weights as below:

$$PF = \frac{W_{payload}}{W_{payload} + W_{empty}}$$

The B_{OA} is the formula for operational availability:

$$B_{OA} = 5 \cdot PF$$

And the final element, B_{AD} is the assembly demonstration bonus:

$$B_{AD} = 15 - 0.1 \cdot t$$

Where:

t is the time in seconds to assemble the aircraft

n is the total number of successful flight rounds flown

m is the total number of flight rounds in the competition

We took the following steps to analyze the scoring. We assumed that the model will weigh minimum 200g and maximum 400g, depending on the configuration of the power plant, batteries, airframe design decisions, and quality of craftsmanship. The last factor accounts for issues resulting accidental damage during construction, precision in machining parts and using/not using optimal amounts of adhesives and resins. The load consists of steel plates fitting into the 1,5”x1,5”x5” cargo box, weighing 173g each. Furthermore, after analyzing the past project and the available design possibilities, we decided that our case will be 13,78” long, as this is the smallest possible case to fit our model while retaining a reasonable amount of parts to assemble. Knowing these initial conditions, we plotted the Rn factor for aircraft weights from 200 g to 400 g, and payload weight for from 1 to 6 load plates. The graph in Fig. 4. tells us, that the heavier the load gets, the less gain in points we achieve with each additional plate, however a heavy aircraft, even when carrying 6 plates, will only achieve a score of a very light aircraft carrying just 3 plates (519 g or 1,44 lbs). This shows how essential it is for the aircraft to have low weight, to achieve

a high Rn score. The B_{OA} factor presents a gain of, on average, 3,15 points per round, while maintaining similar relations between the graphed scenarios.

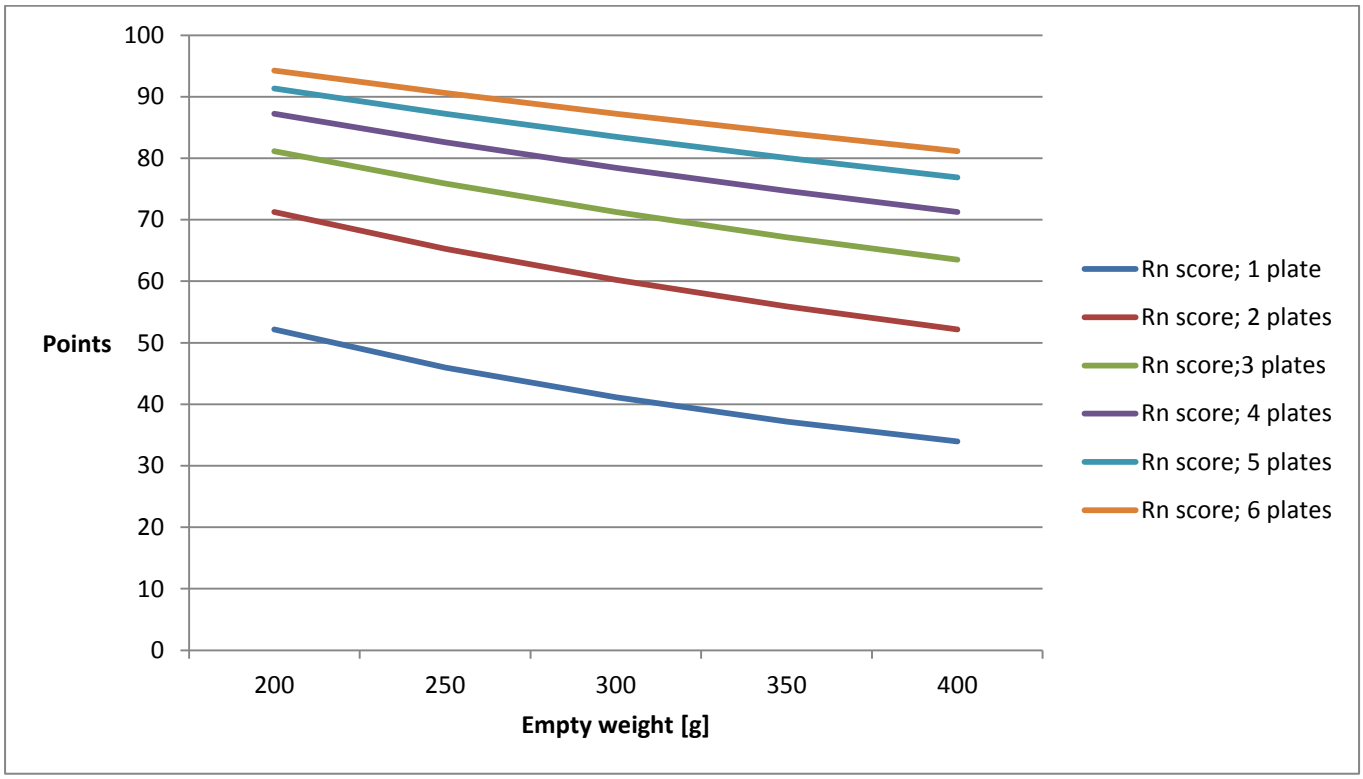


Fig. 4. Score analysis

2.1.2 Translating constraints, mission requirements and scoring into figures of merit for design

Figure of merit	**Weight [%]**	**Additional comments**
Aircraft weight	35	Key role in scoring Flight Round (FR) points
Aircraft max. load	25	Secondary key role in scoring FR points
Aircraft size	15	Considering the previous designs, there is not much room for change, however the L_{AC} factor influences the score for every FR
Historical data	15	This factor reflects the similarity to older designs. It's important because of the successes of the previous airplanes, predictability of the vehicle, and largely influences ease of manufacturing
Ease of manufacturing	5	Influences the aircraft weight, assembly time, ease of repairs in case of structural failures, prevents potential faults in manufacturing

Fast assembly	5	Maximum bonus for fast assembly is 15 points, presuming an unrealistic 0 s. assembly time. Minor importance in comparison with FR points.

Fig. 5. Figures of merit table

While designing the "Pszczola" we kept these figures in mind.

2.1.3. Predicted weather conditions

In order to adapt the aircraft to the local weather conditions and to have figures to base on in calculations, we researched the Florida climate in march. We analyzed data from 2012-2014, the results are shown in the graphs below.

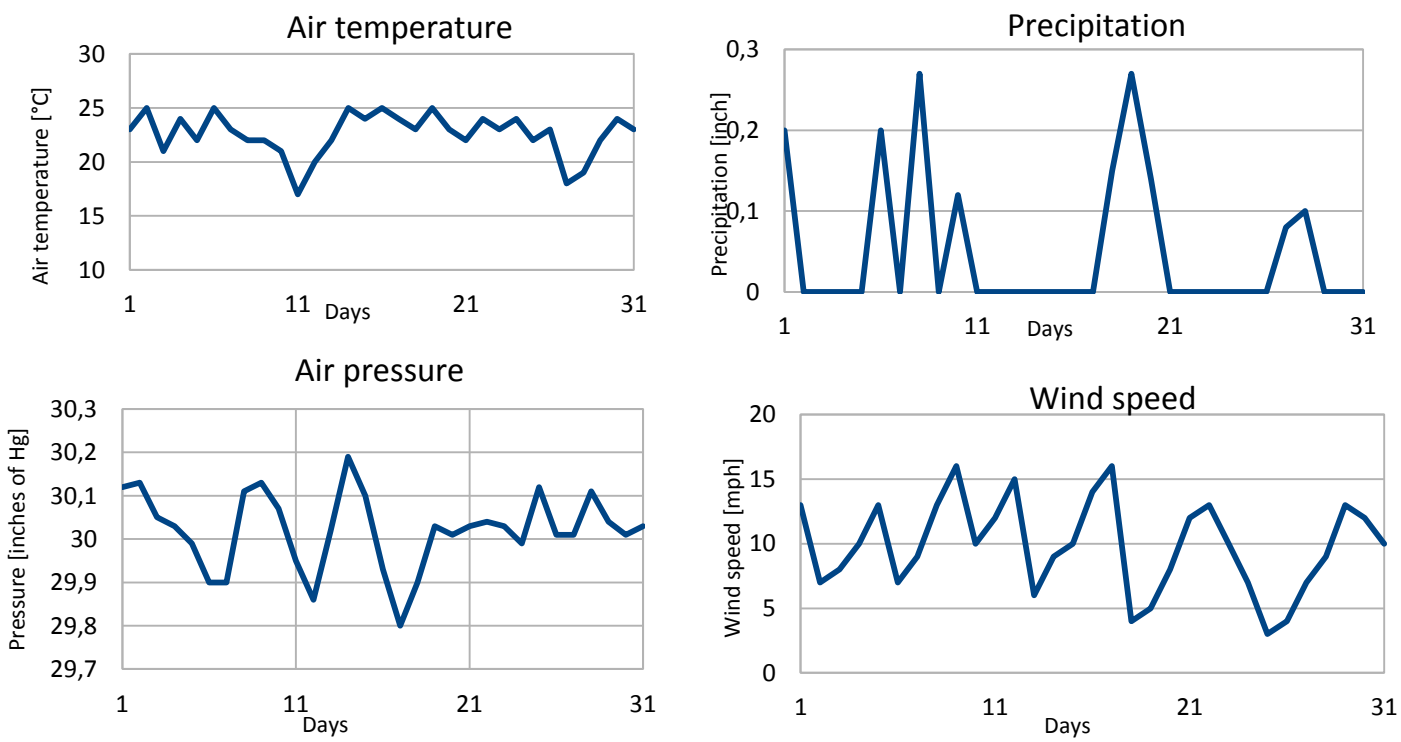


Fig. 6. Weather analysis graphs

The average values are as follows. Temperature: 22.6 [^{o}C], precipitation: 0.05 [in.] Air pressure: 30.03 [in./Hg] (1016.9 [hPa]), wind speed: 9,83 [mph]. Lakeland's elevation of 197 feet above sea level is also taken into account in all the calculations and simulations where it is relevant (Reynold number, power train performance, etc).

2.1.4 Discussion of concepts

Before coming up with the final concept, we considered a number of configurations for wings, motors, control surfaces and cargo bay. The decision making process is shown below.

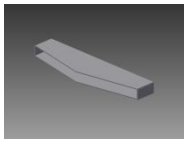
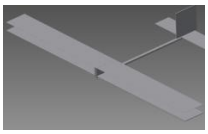
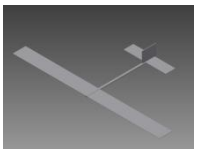
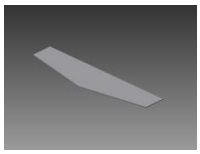
					
Figure of merit	**Weight**	**Bi-wing**	**Biplane**	**Conventional**	**Flying wing**
Mass	35	2	1	5	4
Ease of manufacturing	20	1	2	5	3
Historical data	25	1	1	5	1
Stability and control	20	2	3	4	1
Total	**100 %**	**15,5**	**16**	**48**	**24,5**

Fig. 7. Wing configuration scoring matrix

Although flying wing, Bi-wing, as well as integrated wing-body (not listed) designs are interesting, they do not provide sufficient advantages to win over traditional designs, and pose problems when considering the cargo pod placement, as well as in-flight stability and control, which could be hard to predict, given, that our team has no experience with such designs.

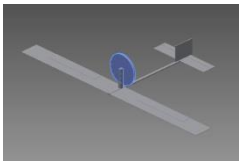
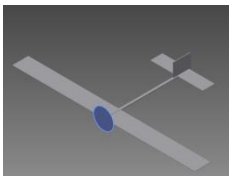
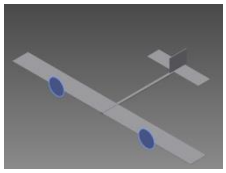
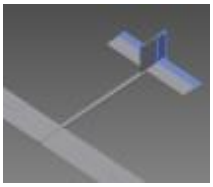
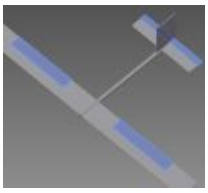
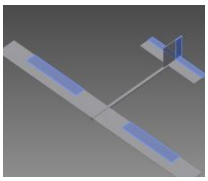
				
Figure of merit	**Weight**	**Pusher**	**Single tractor**	**Twin Tractor**
Mass	30	3	4	2
Ease of manufacturing	20	1	4	1
Power	30	3	3	5
Historical data	20	1	5	1
Total	**100%**	**22**	**39**	**25**

Fig. 8. Motor placement configuration scoring matrix

The Pusher provides similar thrust to a traditional single tractor, while requiring a special motor mount, which increases weight and disturbs the balance of the aircraft by not being in the fuselage's longitudinal axis. The twin tractor provides additional power, but at the cost of substantial mass increase. It would also require significant wing structure and wing-body nodes reinforcement.

					
Figure of merit	**Weight**	**Conventional RE**	**Conventional EA**	**Conventional REA**	**V-tail**
Stability and control	30	4	3	5	4
Mass	20	4	3	1	4
Ease of manufacturing	20	4	3	3	4
Historical data	20	5	4	5	4
Drag	10	4	3	3	4

Total	**100%**	**42**	**32**	**36**	**40**

Fig. 9. Control surfaces scoring matrix

The conventional rudder + elevator solution was picked as the main design decision, however we also seriously considered the runner-up, the V-tail, also known as Rudlicki Empennage, because of its low drag, good steering capabilities, and our previous experiences with it. We have chosen to build prototypes with both kinds of control surfaces. The flight test has proven the conventional design to provide better flight control in our particular case. The conventional REA solution would provide superior maneuvering capabilities, but our experience with the antecedent model, the "Big Bang" tells us that RE is sufficient, while maintaining lower weight and drag.

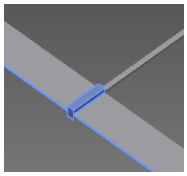
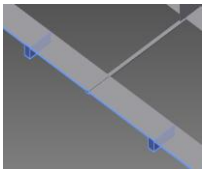
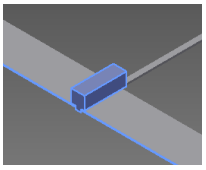
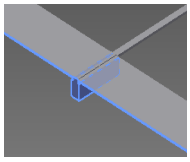
					
Figure of merit	**Weight**	**Integrated**	**Double pods**	**High pod**	**Low pod**
Mass	30	4	2	4	3
Ease of manufacturing	20	1	3	4	4
Historical data	20	3	2	1	5
Drag	20	4	2	3	3
Stability	10	4	2	1	5
Total	**100%**	**32**	**22**	**29**	**38**

Fig.10. Cargo pod location scoring matrix

Low pod provides a lower placement of the aircraft's center of gravity, adding to stability; it can function as a skid, eliminating the need for landing gear and is a lot easier to integrate with the airframe than the double pods or the integrated pod. It can also be loaded and unloaded quickly.

2.1.5 Discussion of previous designs

The antecedent vehicle, dubbed “Big Bang”, while successful on its own, is a part of a family of UAVs designed for 6 earlier SAE Aero Design competition editions, some of which, including the “parent” of the “Big Bang”, were quite successful. “Big Bang” was stable, provided good control to the pilot, and had very good empty weight and max. load weight, which provided it with the 3rd placement in the Payload Fraction category. His “parent” design achieved 2nd place, while being heavier and sturdier. It is reasonable to think, that basing on a similar design, while fine-tuning the balance between the load weight and the empty weight might be the right approach in our case. Nonetheless, as shown in point 2.1.4, we took under consideration various other configurations and possibilities in the designing process.

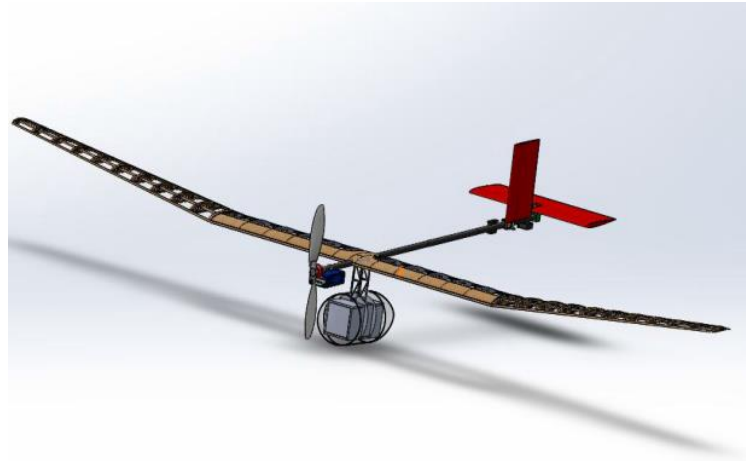


Fig. 11. The “Big Bang” drawing

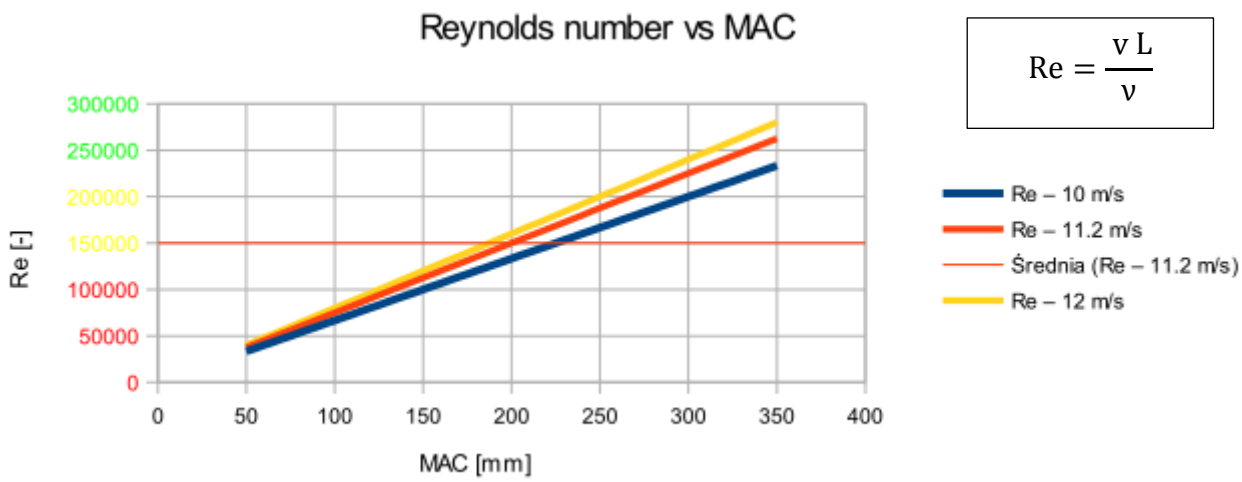
2.1.6 Overall configuration choice

Considering our main points of merit, derived from the constraints and scoring predictions, we decided for a conventional wing, single tractor with a low pod and control surfaces comprising of a conventional rudder and elevator. We decided against landing gear, as the pod may take its function.

2.2 Preliminary design

2.2.1 Wing planform design

For designing the wing planform, we had to perform some initial calculations and assumptions. We assumed a **Reynolds number (Re)** of 100,000 or higher would be optimal. From that we derived a



$$Re = \frac{v\,L}{\nu}$$

Fig. 12. Re vs MAC graph

Mean Aerodynamic Chord (MAC), which would allow us to achieve that number in the Florida climate, at the average airspeed of 10m/s. As you can see in Fig. 12., the 150 mm MAC fits these conditions without being oversized, so the wing cord was decided to be at 150 mm.

Because we are following the example of the 2013 model, we agreed to have a similar wing span, decreased by 78mm for a perfect fit in the tube casing. Therefore, the wingspan of "Pszczola" is 1415 mm. This gives us an **Aspect Ratio** (AR) of 9.43.

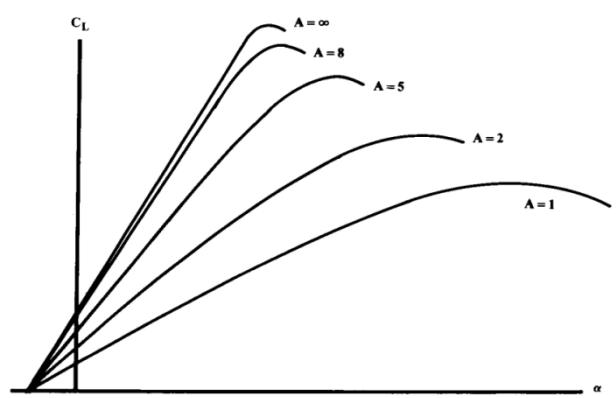


Fig. 13. AR vs CL and angle of attack; after: Raymer, p. 50

No taper was chosen because of ease of manufacturing, which also adds the benefit of not compromising Re at the wingtips.

Following our previous experiences, we added a dihedral angle of 6° on the wing tips, to prevent dutch roll and improve spiral stability.

Incidence angle was set to 0° to minimize drag.

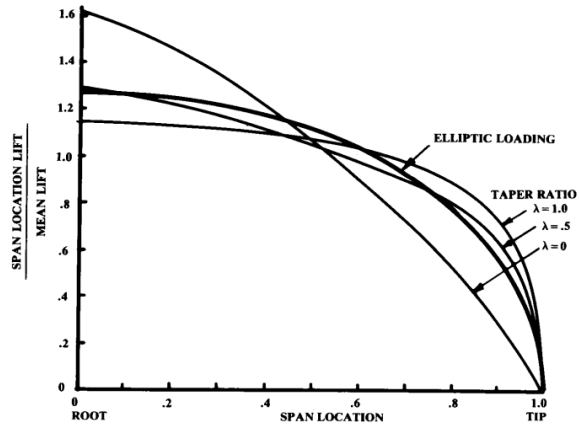


Fig. 14. Taper ratio vs Lift; after: Raymer p. 55

2.2.2 Airfoil selection

We were looking for airfoils that would provide high lift at low Reynolds numbers. Initially we limited our choices to three airfoils: S1223, SG6042 and SG6043. However, the strong camber of the S1223 made it difficult to fit the wings with the rest of the aircraft inside the carrying case, without shortening the MAC, which, in turn would diminish the lift force of the wing. Furthermore, the trailing edge of S1223 is very thin, which makes it prone to damage and is difficult to manufacture. These drawbacks led us to decide against using this airfoil. The remaining two were constructed and flight tested. The airfoils are presented below.

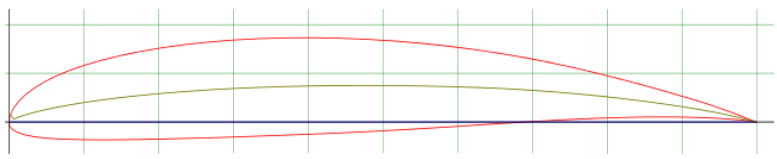


Fig. 15. SG6042 airfoil

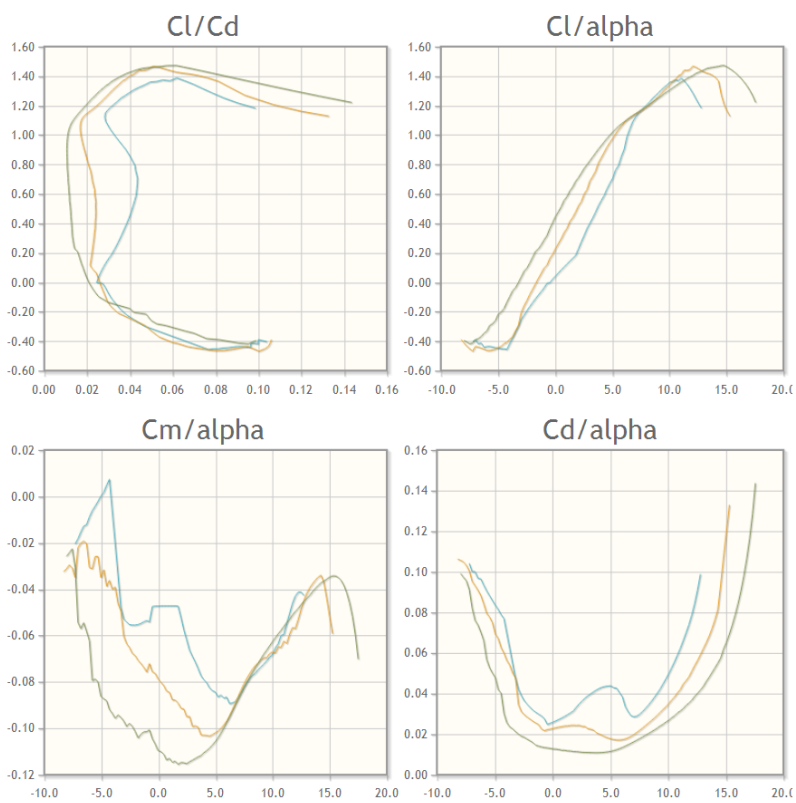


Fig. 16. SG6042 characteristics from Xfoil. Blue: Re=50,000, yellow: Re=100,000, green: Re=200,000

The airfoils were chosen by analyzing their key characteristics: lift to drag ratio (Cl/Cd), dependency of lift coefficient on angle of attack (Cl/alpha), pitching moment coefficient to angle of

attack ratio (Cm/alpha) and drag coefficient to angle of attack ratio (Cd/alpha). The graphs shown to the left and below come from a simulation performed in Xfoil software. Each diagram plots these characteristics for 3 variants of Re (50,000; 100,000; 200,000), as our flights should oscillate around Re=100,000, this plot is most important for our choice process.

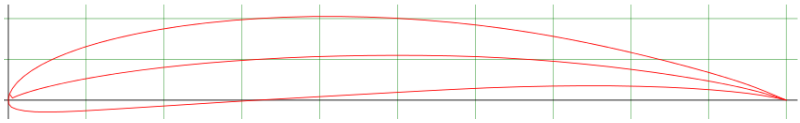


Fig. 17. SG6043 airfoil

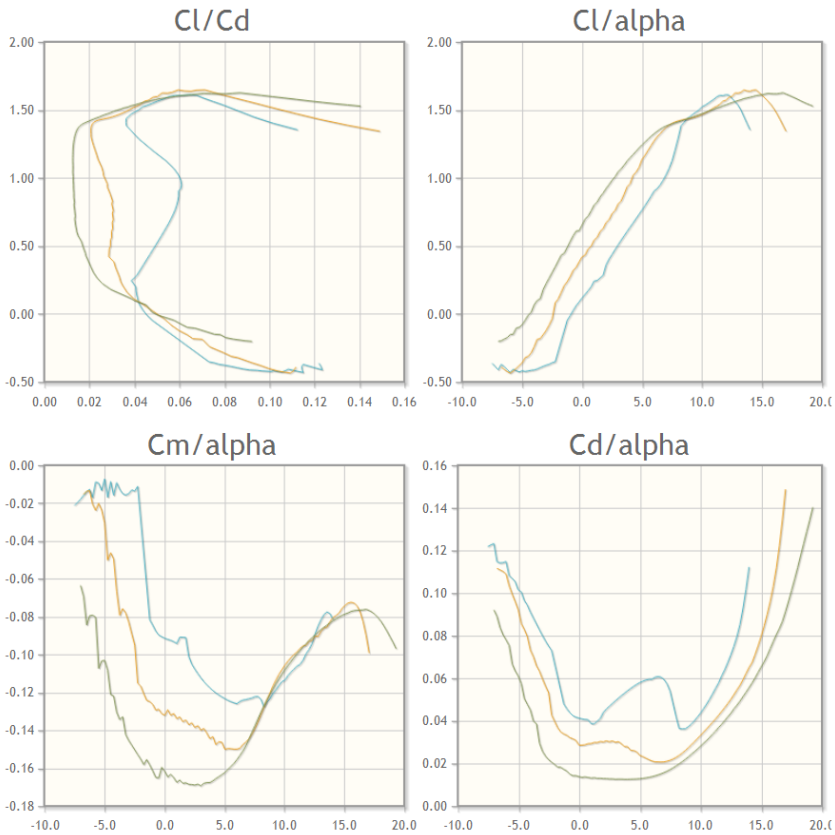


Fig. 18. SG6043 characteristics from Xfoil. Blue: Re=50,000, yellow: Re=100,000, green: Re=200,000

At $\alpha = 0^o$ the SG6043 has a lift coefficient of almost 0.5, and the drag coefficient that equals 0.03, while at the same angle the SG6042 has Cl = 0.2, with a relatively small decrease in drag (Cd = 0.021). This indicates that the SG6043 might provide better performance for lifting heavy cargo with limited available power, as is the case in this competition.

However the conclusion of the test flights was that the SG6042 provided better handling, higher airspeeds, and, in consequence, better overall performance, so the final decision was to go with the SG6042 airfoil.

2.2.3 Airframe and fuselage design

The fuselage is a carbon tube, with the engine attached in the front section, the wing and load tower assembly in the middle section and the empennage in the aft section, as shown in Fig. 19. To make it fit into the carrying case, the tube is divided into 3 parts, each 350 mm long, which are connected together via friction fit. The load support tower is a CNC milled framework, made from sheets of glass-epoxy composite. From the structural point of view, the tower is one of the most crucial elements of this aircraft, because it supports the wings, connects them with the fuselage, carries the full weight of the cargo pod, and, since the pod is also a landing skid, it has to withstand the touch-down forces. The structure proved to be prone to structural failures in the past, so we analyzed the old tower design in Siemens FEMAP (Fig. 20.) and optimized the framework by thinning or thickening the frame parts according to shear stress. We also added an aerodynamic casing for the tower, as seen in Fig. 19. According to our flow analysis in Autodesk Flow, this lowered the Cd of the tower by 0.05

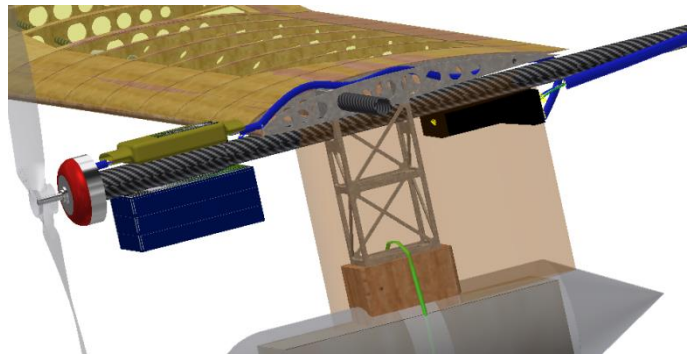


Fig. 19. Wing mount and tower view

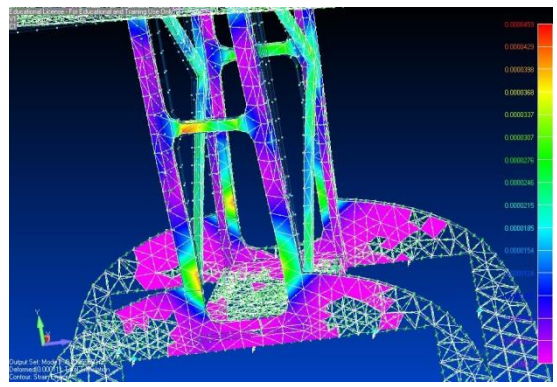


Fig. 20. FEM stress analysis in FEMAP

2.2.4 Tail design

The tail is arranged in a conventional rudder-elevator layout. Its length is a tradeoff between weight, proper control and ability to fit in a small case. After assembly, the fuselage boom is 1065.35 mm long. To achieve good stability, we iterated the control surfaces to achieve favorable tail volume coefficients, calculated using the formulas below:

Horizontal tail surface	Vertical tail surface
$S_h = \frac{V_h \cdot S_{ref} \cdot MAC}{L_h} = 0{,}0285\ m^2$	$S_v = \frac{V_v \cdot S_{ref} \cdot MAC}{L_v} = 0{,}014\ m^2$

Where:

- V_h, V_v – tail volume coefficients
- S_{ref} – wing area
- L_h, L_v – control surface torque arm lenght

2.2.5 Payload bay

This year's SAE constraints for the cargo box were 1.5 x 1.5 x 5 [in.]. We applied a simple approach of creating the box out of plywood, similarly to the 2013 design. We also added a similar aerodynamic PET casing.

2.2.6 Propulsion

Based on our previous positive experiences with good thrust parameters at low operating speeds, we decided to use a 8x4 propeller from GWS. As for engine choice we compared a number of candidates suggested by MotoCalc software. The strongest candidates were Welgard C2024-25, BPA2208-12 and E-Max GT2205/22. Their simulated performances were rather similar, however in the end we decided to use E-Max GT2205/22, as we have used many E-Max motors in the past and we value them for their reliability and consistent performance. Below we present a simulation data sheet from MotoCalc, which includes static and dynamic thrust.

As for the power source, it was decided that we should use Turnigy Nano Cell 3S 4mAh battery, to conserve weight.

AirSpd (mph)	Drag (oz)	Lift (oz)	Batt Amps	Motor Amps	Motor Volts	Input (W)	Loss (W)	MGbOut (W)	MotGb Ef(%)	Shaft Ef(%)	Prop RPM	Thrust (oz)	PSpd (mph)	Prop Ef(%)	Total Ef(%)	Time (m:s)
0,0	0,0	0,0	12,9	12,9	6,7	86,1	24,3	61,7	71,7	64,5	9162	13,5	34,7	0,0	0,0	3:23
1,0	0,0	0,1	12,9	12,9	6,7	86,1	24,3	61,7	71,7	64,5	9162	13,3	33,7	2,7	1,7	3:23
2,0	0,0	0,3	12,9	12,9	6,7	86,1	24,3	61,7	71,7	64,5	9162	13,1	32,7	5,3	3,4	3:23
3,0	0,0	0,6	12,9	12,9	6,7	86,1	24,4	61,7	71,7	64,5	9160	12,9	31,7	7,8	5,0	3:23
4,0	0,1	1,1	12,9	12,9	6,7	86,1	24,4	61,7	71,7	64,5	9157	12,8	30,7	10,2	6,6	3:23
5,0	0,1	1,6	12,9	12,9	6,7	86,1	24,4	61,7	71,7	64,5	9155	12,6	29,7	12,6	8,1	3:23
6,0	0,2	2,4	12,9	12,9	6,7	86,2	24,4	61,8	71,7	64,5	9152	12,4	28,7	14,9	9,6	3:23
7,0	0,2	3,2	13,0	13,0	6,7	86,2	24,4	61,8	71,6	64,5	9150	12,2	27,7	17,1	11,0	3:23
8,0	0,3	4,2	13,0	13,0	6,7	86,2	24,5	61,8	71,6	64,4	9149	12,0	26,7	19,3	12,4	3:23
9,0	0,4	5,3	13,0	13,0	6,7	86,2	24,4	61,8	71,6	64,4	9149	11,8	25,7	21,3	13,7	3:23
10,0	0,5	6,6	12,9	12,9	6,7	86,1	24,4	61,7	71,7	64,5	9160	11,6	24,7	23,3	15,0	3:23
11,0	0,6	8,0	12,9	12,9	6,7	85,8	24,2	61,6	71,8	64,6	9178	11,3	23,8	25,1	16,2	3:24
12,0	0,7	9,5	12,8	12,8	6,7	85,4	23,9	61,5	72,0	64,8	9213	11,1	22,9	26,9	17,4	3:25
13,0	0,8	11,1	12,7	12,7	6,7	84,7	23,5	61,2	72,3	65,1	9259	10,8	22,1	28,5	18,5	3:27
14,0	1,0	12,9	12,6	12,6	6,7	84,0	23,0	60,9	72,6	65,5	9316	10,5	21,3	30,0	19,6	3:29
15,0	1,1	14,8	12,4	12,4	6,7	83,0	22,5	60,6	72,9	65,9	9382	10,2	20,5	31,4	20,7	3:32
16,0	1,3	16,9	12,2	12,2	6,7	82,0	21,9	60,1	73,3	66,4	9458	9,9	19,8	32,7	21,7	3:35
17,0	1,4	19,1	12,0	12,0	6,7	80,7	21,2	59,5	73,8	66,9	9545	9,6	19,2	33,9	22,7	3:39
18,0	1,6	21,4	11,8	11,8	6,7	79,2	20,4	58,8	74,3	67,5	9647	9,2	18,5	35,0	23,6	3:43
19,0	1,8	23,8	11,5	11,5	6,7	77,4	19,5	57,9	74,9	68,2	9763	8,9	18,0	36,0	24,5	3:49
20,0	2,0	26,4	11,2	11,2	6,8	75,4	18,5	56,9	75,5	68,9	9892	8,5	17,5	36,9	25,4	3:56
21,0	2,2	29,1	10,8	10,8	6,8	73,3	17,5	55,9	76,2	69,8	10039	8,1	17,0	37,7	26,3	4:03
22,0	2,4	31,9	10,4	10,4	6,8	70,8	16,4	54,4	76,8	70,6	10187	7,7	16,6	38,4	27,1	4:13
23,0	2,7	34,9	10,0	10,0	6,8	68,0	15,4	52,6	77,4	71,4	10344	7,2	16,2	39,1	27,9	4:24
24,0	2,9	38,0	9,5	9,5	6,9	64,9	14,3	50,6	78,0	72,2	10510	6,8	15,8	39,7	28,7	4:37
25,0	3,1	41,2	9,0	9,0	6,9	61,7	13,3	48,4	78,5	73,0	10683	6,3	15,5	40,3	29,4	4:53
26,0	3,4	44,6	8,4	8,4	6,9	58,1	12,2	45,9	78,9	73,8	10863	5,8	15,1	40,8	30,1	5:13
27,0	3,7	48,1	7,8	7,8	7,0	54,4	11,3	43,1	79,3	74,5	11049	5,3	14,9	41,2	30,7	5:36
28,0	3,9	51,7	7,2	7,2	7,0	50,4	10,3	40,0	79,5	75,0	11241	4,8	14,6	41,6	31,2	6:05
29,0	4,2	55,4	6,6	6,6	7,0	46,1	9,5	36,6	79,5	75,4	11439	4,3	14,3	42,0	31,7	6:40

Motor performance calculations take ambient temperature and heating effects into account.

Color Key: Propeller Stalled | Stall Speed @ Clmax=1,3 | Level Flight @ Clopt=0,86 | Level Flight @ Cl=0,78

Fig. 21. Motor data from MotoCalc

2.3 Detail design, analysis and review

2.3.1 Wing structure

The wing is composed of ribs, caissons, struts, as well as attack and trailing surfaces – all of which are made of 1mm thick balsa wood. All the elements are hand carved, except the airfoil-shaped ribs which are CNC cut. Furthermore, the wing is strengthened with a carbon fiber tube. The centerwing has an 8mm thick

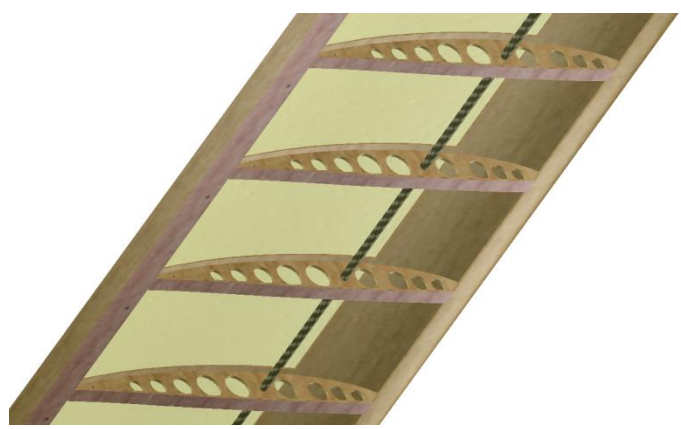


Fig. 22. Wing structure

one, and the wingtips, which carry less stress, has a thinner one, with the diagonal diminishing from 3mm at the centerwing side to 2mm at the tip side.

After initial structural problems and failure at the drop test, the centerwing-to-tower connection was further reinforced with an aluminum ϕ 8mm tube and a block of balsa wood. Wing and control surfaces have been covered with a heat-shrinking foil for minimal surface roughness.

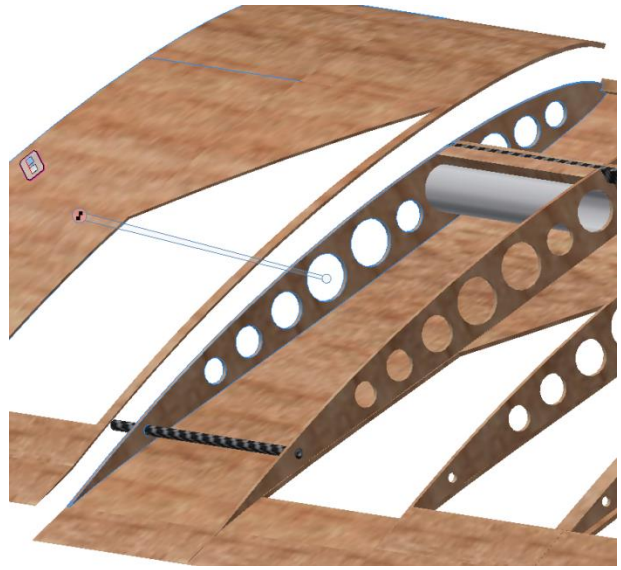


Fig. 23. Centerwing mount structure

2.3.2 Computational analysis and review

2.3.2.1. Normal mode oscillation frequencies in Siemens FEMAP

The 2013 and 2015 aircrafts' central elements were tested for frequency responses using the Eigenvalue analysis. Due to the fact that balsa is a soft wood, with good damping qualities, none of the resonant frequencies occurred to be dangerous for the integrity of the aircraft.

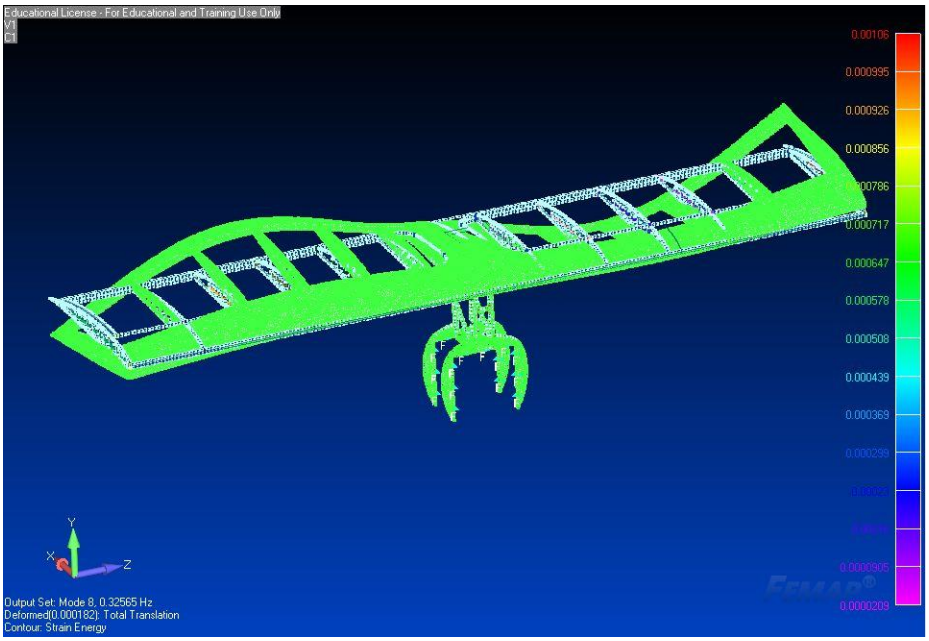


Fig. 24. Normal mode oscillation example, FEMAP

2.3.2.2 Computational Fluid Dynamics analysis in Autodesk Flow

Various iterations and variants of the design were tested in a digital wind tunnel, which let us pick solutions that would induce the smallest drag forces. Shown below as an example, are airspeed and pressure distributions in the flow around the central tower: without a casing, with a partial casing and with full aerodynamic casing. These analyses also helped estimate the drag coefficients for different parts of the aircraft.

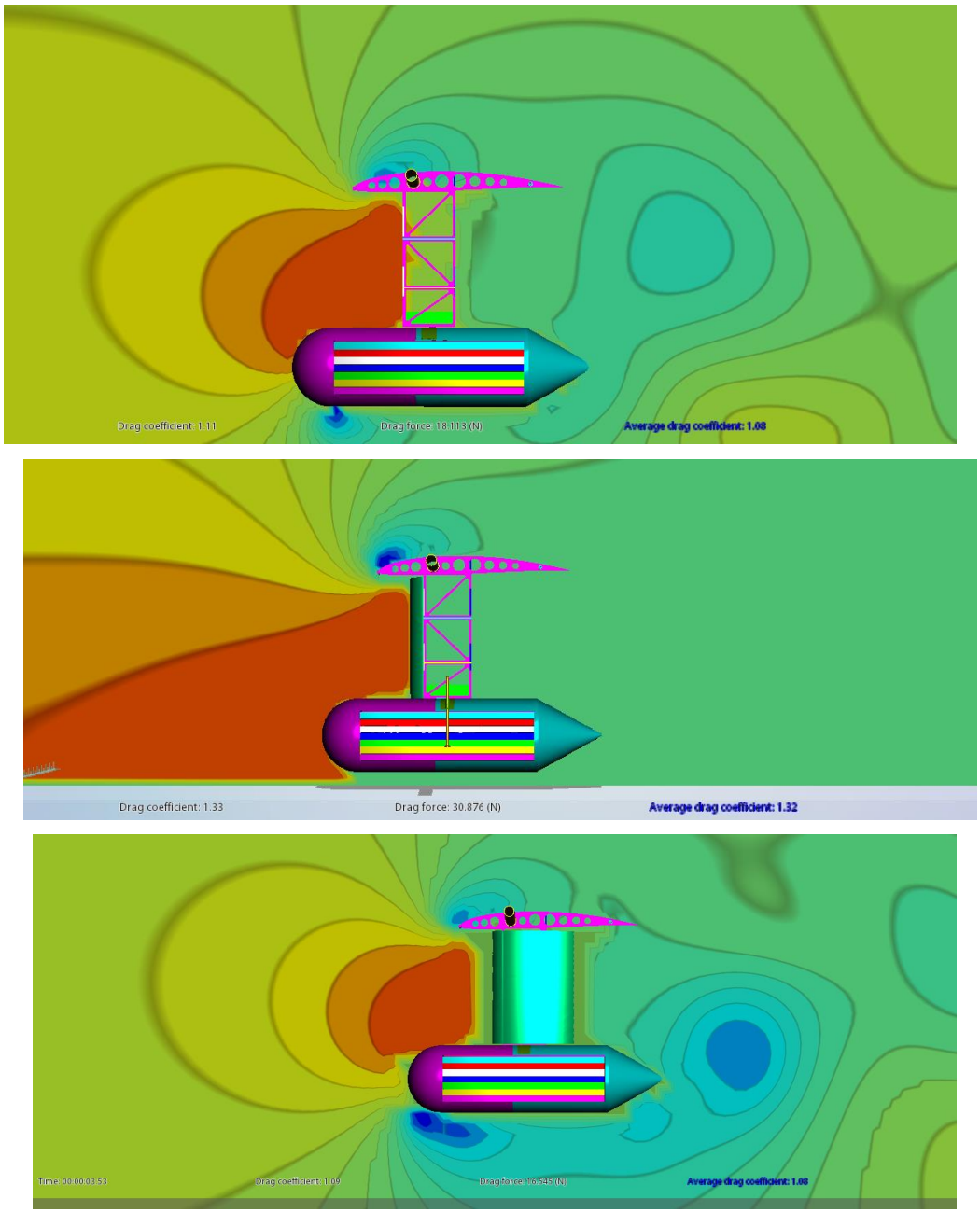


Fig. 25. Aerodynamic flow around the tower; Autodesk Flow

2.3.3 Test flights

Three different prototypes were built: SG6042 w/ conventional empennage, SG6043 w/conventional empennage and SG6042 w/ V-tail controls. The V-tail occurred to heavily lack in stability and ease of control, while the conventional SG6043 was stable, but did not perform well in terms of lift and stall speed. After the tests, we discovered structural failures in the lower part of the tower framework (glass-epoxy composite strut failure) and the wing mount, which we later addressed while redesigning these critical points.

Fig. 26. „Pszczola" during test flight

2.3.4 Drop testing

Each prototype was dropped in a level flight position, from a height of 30 cm. Failures at centerwing mount which occurred during the test were later addressed in the design by reinforcing the mount with an aluminum tube, 8 mm in diameter, and a balsa wood block supporting the said tube.

2.3.4 Virtual fit test

Before the model was manufactured, we made sure that it will fit without collision in the case of assumed length. The test was a success.

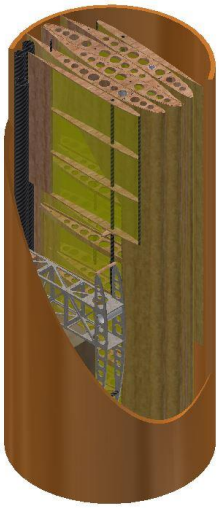


Fig. 27. Virtual fit test; Autodesk Inventor

3. Calculations

3.1 Stability and control

Stability analysis was performed in XFRL5 software. Stability and control derivatives as well as normal mode values are presented below.

Lateral derivatives				Longitudinal derivatives			
Yv=	-0.099962	CYb=	-0.10513	Xu=	-0.0091539	Cxu=	-0.0096272
Yp=	-0.080328	CYp=	-0.1408	Xw=	0.16241	Cxa=	0.17081
Yr=	0.051013	CYr=	0.089418	Zu=	-0.52097	Czu=	-0.0024012
Lv=	-0.089216	Clb=	-0.078191	Zw=	-4.1022	CLa=	4.3143
Lp=	-0.35489	Clp=	-0.51839	Zq=	-0.15543	CLq=	2.1795
Lr=	0.060716	Clr=	0.088688	Mu=	0.0002526	Cmu=	0.0017711
Nv=	0.022855	Cnb=	0.02003	Mw=	0.11739	Cma=	0.82303
Np=	-0.039967	Cnp=	-0.05838	Mq=	-0.083829	Cmq=	-7.8368
Nr=	-0.015538	Cnr=	-0.022697	Neutral Point position=		0.07826 m	

Control derivatives			
Xde=	0.63718	CXde=	0.0777
Yde=	0.066647	CYde=	0.0081273
Zde=	-34.464	CZde=	-4.2027
Lde=	-0.054111	CLde=	-0.0054988
Mde=	0.8769	CMde=	0.71289
Nde=	-0.15465	CNde=	-0.015716

___Longitudinal modes____

Eigenvalue:	-24.43+	0i	|	-2.058+	-1.792i	|	-2.058+	1.792i	|	1.525+	0i
Eigenvector:	1+	0i	|	1+	0i	|	1+	0i	|	1+	0i
	-63.34+	0i	|	-0.1623+	-0.3525i	|	-0.1623+	0.3525i	|	-0.2263+	0i
	51.62+	0i	|	-0.1175+	-0.6706i	|	-0.1175+	0.6706i	|	-0.2683+	0i
	-2.113+	0i	|	0.1939+	0.1571i	|	0.1939+	-0.1571i	|	-0.176+	0i

___Lateral modes____

Eigenvalue:	-29.57+	0i	|	-0.5761+	-3.955i	|	-0.5761+	3.955i	|	0.0001145+	0i
Eigenvector:	1+	0i	|	1+	0i	|	1+	0i	|	1+	0i
	21.6+	0i	|	-0.2547+	0.03375i	|	-0.2547+	-0.03375i	|	0.0001492+	0i
	1.709+	0i	|	0.02804+	0.3943i	|	0.02804+	-0.3943i	|	1.47+	0i
	-0.7305+	0i	|	0.0008304+	-0.06428i	|	0.0008304+	0.06428i	|	1.304+	0i

Fig. 29. Stability and control data from XFLR5

3.2 Weight buildup

3.2.1 The weight buildup estimated and final values are presented in the table below, along with a commentary describing the discrepancies.

Part name	Estimated weight [g]	Final weight [g]	Commentary
Left centerwing	20	21	The differences for these airframe elements derives from varying balsa wood specific weight. The amount of glue and/or epoxy resin also played a role in creating discrepancies.
Right centerwing	20	24	
Left wingtip	12	16	
Right wing tip	12	14	
Servo (1 pc.)	5x2	5x2 = 10	
Rudder	4	5	The difference is caused by an unforeseen necessity to add brackets to the ribs, in crearing the rigidity od the rudder, by giving it a T shape. This was to prevent flanging of this element while covering it with heat-shrinking foil.
Fuselage tube	35	38	
Wires	10	12	
Elevator	8	9	The same as with ruder.
Electric motor	41	41	Data for these elements were taken from catalogues.
Battery	77	77	
Battery	39	39	
Small propeller	9	9	
Big propeller	10	10	
Regulator	28	28	
Cargo bay support	18	21	Difference is caused by a necessity to use more epoxy resin and carbon fiber, in order to strengthen the construction after it failed at a drop-test.
Aerodynamic shield for cargo box	25	19	The estimate was made based on the previous model, which had a bigger cargo box.
Cargo box	25	30	Initial estimate of the necessary thickness of plywood was an underestimate. Thicker material was used to account for the heavy load.

Fig. 30. Weight buildup table

3.2.2 Methods of weight relief

Choosing light and durable materials for the parts was the first step in maintaining the lowest possible weight. Selecting particular pieces of materials with the lowest specific weight was the second step. Every sheet of balsa wood that we used, was weighed, so that we could use the lightest sheets for the biggest surfaces. Surprisingly, the difference between two 1m x 0,1m x 0,001m sheets can reach as much as 100% (12g vs 22g), sometimes even more.

3.3 Drag analysis

We performed drag analysis in XFLR5. The drag and lift coefficients for three different angles of attack are shown in Fig. 31, 32, and 33, below.

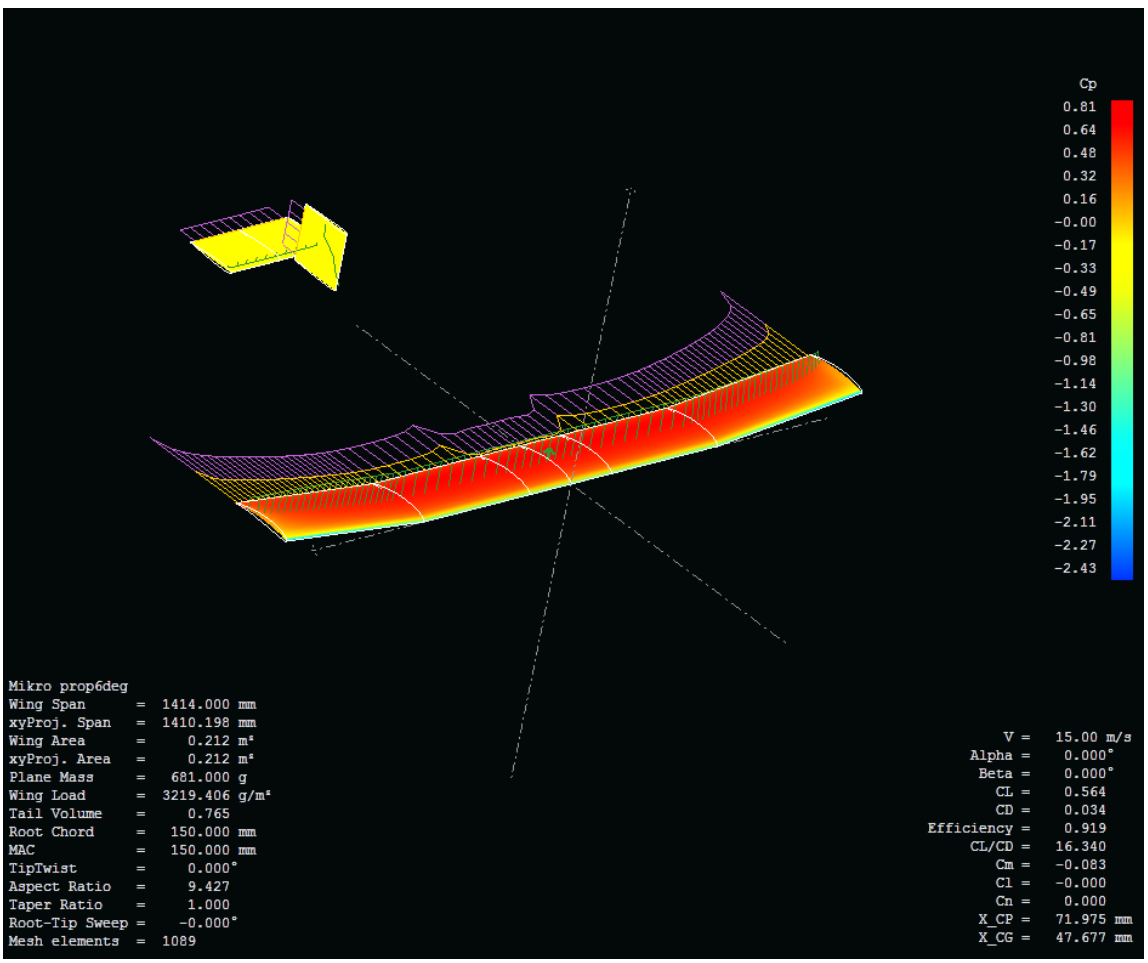


Fig. 31. XFLR5 Drag/lift analysis and data for $\alpha = 0^0$

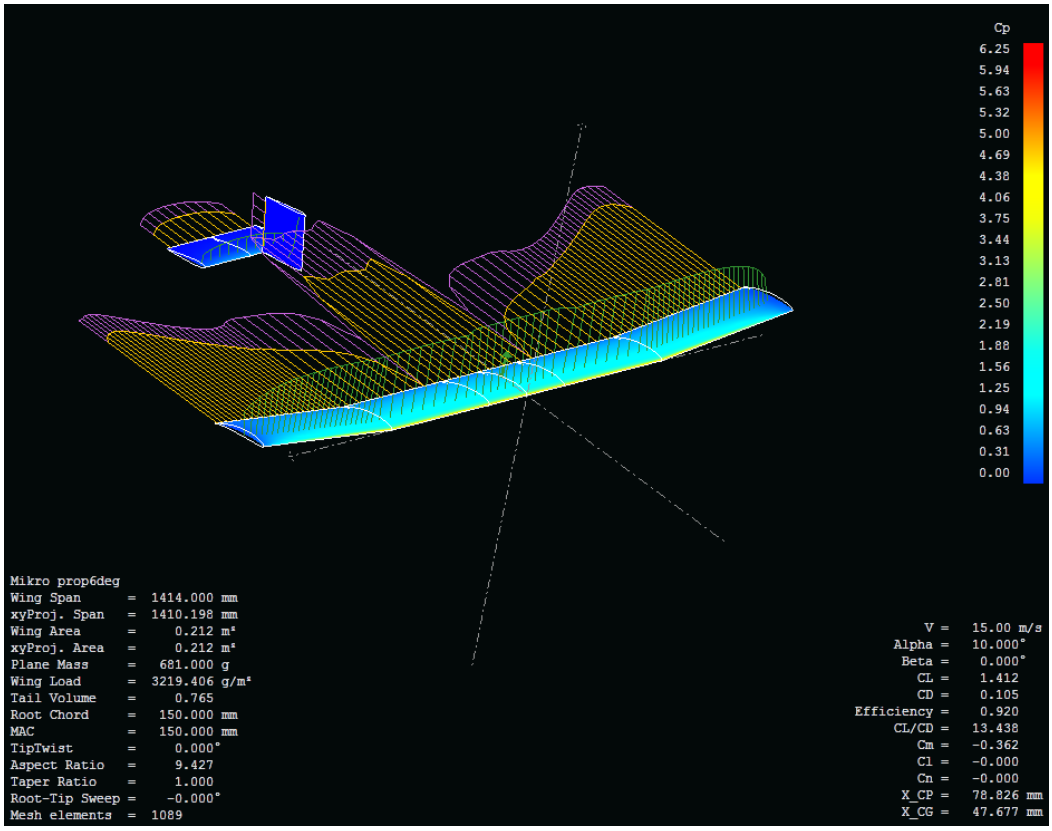


Fig. 32. XFLR5drag/lift analysis for $\alpha = 10^0$

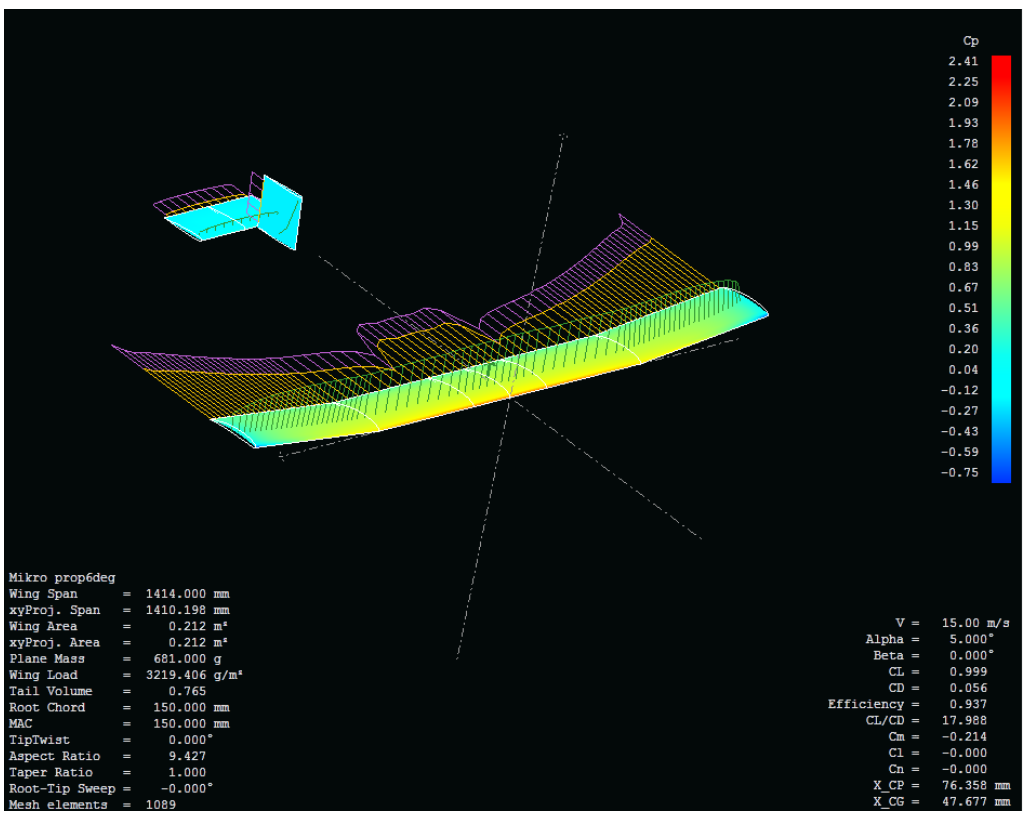


Fig. 33. XFLR5drag/lift analysis for $\alpha = 5^0$

4. Innovations

4.1 Finite elements method of structural analysis

The following methods of design anlysis were introduced this year for the first time, and have proven to be very helpful in optimizing the aircraft.

- Stress analysis using Siemens FEMAP v11.1.2 – as described in the previous design descriptions.
- Normal mode resonant frequency response using Siemens FEMAP v11.1.2 – as above.

4.2 Other

- Engine static and dynamic thrust and power efficiency analysis in MotoCalc software.
- Aerodynamic flow was tested in Autodesk Flow software.
- The MotoCalc data was verified using a newly built test bed, suitable for both Micro and Regular class motors. The data was collected by a KM-102 shear stress gauge, through a laboratory grade Agilent Vee Pro 22 bit A/D converter/meter and Agilent Vee Pro software.

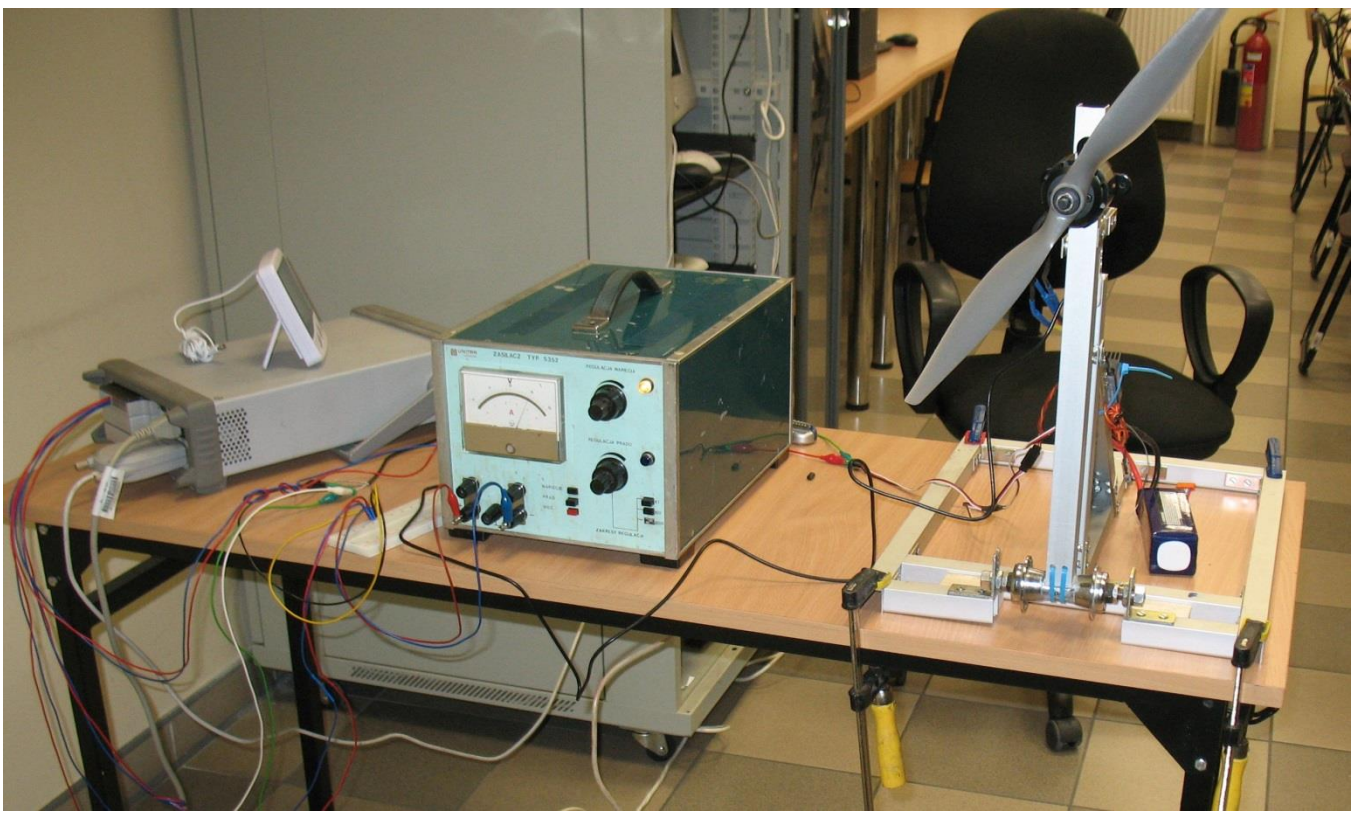


Fig. 34. Laboratory setup for the engine test stand. From the left: Agilent Vee A/D converter/meter, stabilized power supply, motor test bed.

5. References

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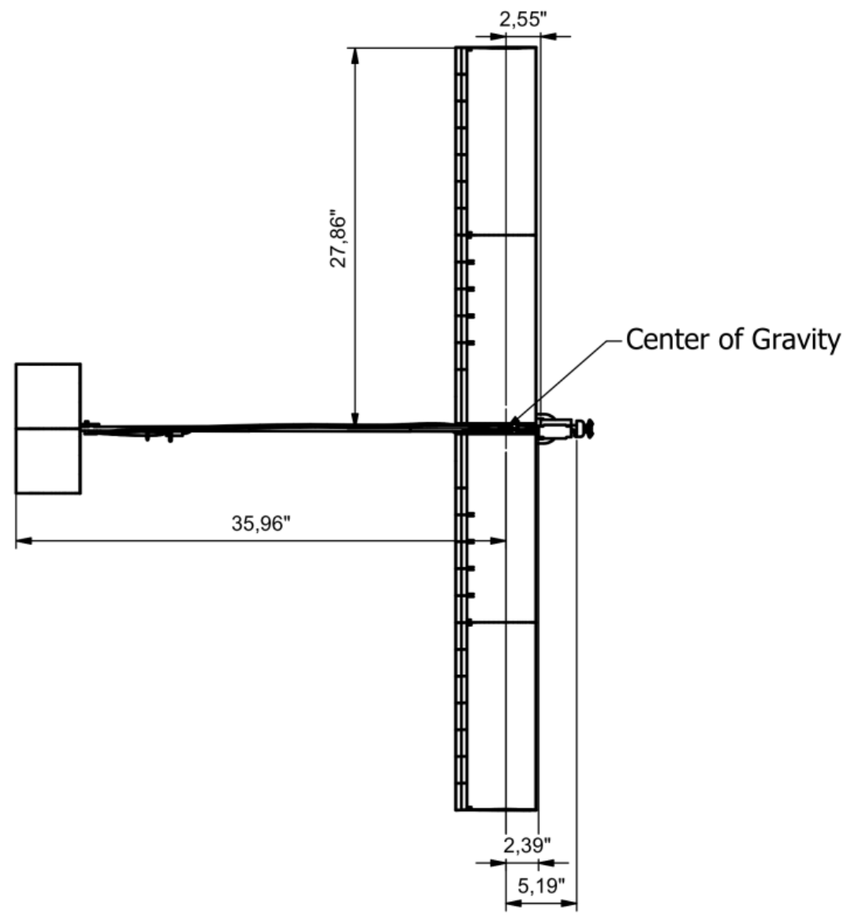
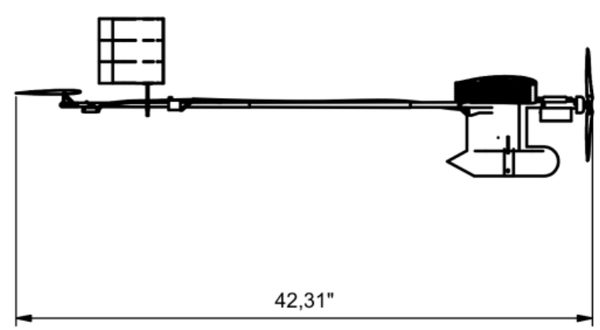
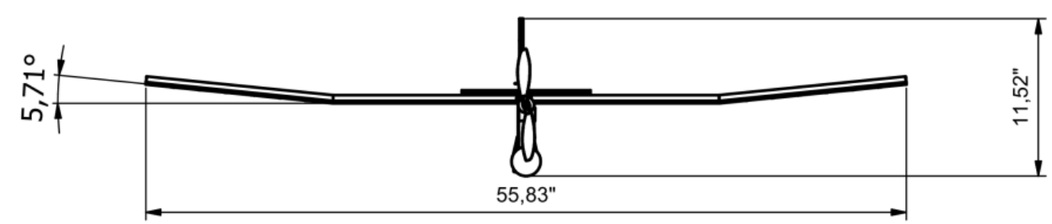
Part name	Weight [g]	Distance form center of gravity ["]	The moment of the center of gravity [Nm]
Engine: EMAX, CF2812	39	5,19	50,41
Battery: nano-tech 850mAh 45-90C	39	2,55	24,77
Engine Regulator Nastic 25A	28	2,39	16,68
Empty model	336	-	-



Payload fraction = (-0.00022*(Density Alt)+4)/(-0.00022*(Density Alt +4.8)





Zaprojektowany przez	Sprawdzony przez	Zatwierdzony przez	Data	Team nr	Data	format:
Maciej Haufa	—	—		#328	2015-01-26	11x17"

Poznan University of Technology	"PSZCZOŁA"		
	Micro Class	Wydanie v 1.0	Arkusz 1 / 1